

TRADEXPERT AN AI-POWERED STOCK MARKET BUDDY

*Project report submitted
For Review (Minor Project) in fulfillment of
The requirement for the degree of*

**Bachelor of Technology
(Computer Science and Engineering)**

By

Vatsa joshi - 22000420

Het Patel - 22000984

Mistry khushal – 22001007

Raj mistry - 22000995

Guided By: Prof. Ninad Bhavsar



**Department of Computer Science and Engineering
School of Engineering and Technology
Navrachana University, Vadodara
May 2022**

CERTIFICATE

This is to certify that the project report entitled “**TradeXpert an ai-powered stock market buddy**” submitted by “**Vatsa Joshi, Het Patel, Mistry Khushal, Raj Mistry**” to School of Engineering and Technology (SET) of Navrachana University Vadodara, in partial fulfillment for the award of the degree of B. Tech in Computer Science and Engineering (CSE) department. This report is a bona fide record work that has been carried out under my supervision during academic year **2024-25**. The contents of this report, in full or in parts, have not been submitted to any other Institution or University for the award of any degree or diploma.

Prof. Ninad Bhavsar

Guide and Assistant Professor
Computer Science and Engineering
School of Engineering and Technology
Navrachana University, Vadodara

Prof. Ninad Bhavsar

Program Chair and Assistant Professor
Computer Science and Engineering
School of Engineering and Technology
Navrachana University, Vadodara

Dr. Ashish Jani

Head and Professor
CSE, IT, BCA and BSc DS
School of Engineering and Technology
Navrachana University, Vadodara

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Name of the student	Student ID	Signature
Vatsa Joshi	22000420	
Het Patel	22000984	
Mistry Khushal	22001007	
Raj Mistry	22000995	

Date: _____

Acknowledgment

Every project big or small is successful largely due to the effort of a number of wonderful people who have always given their valuable advice or lent a helping hand. We sincerely appreciate the inspiration, support and guidance of all those people who have been instrumental in making this project a success. We feel deeply honored in expressing our sincere thanks to our Internal Project Guide **Prof. Ninad Bhavsar** for providing valuable insights leading to the successful completion of the project.

We are extremely grateful to **Prof Ashish Jani**, (Head- CSE, IT BSc-DS, BCA) and **Prof Ninad Bhavsar** (Program Chair- CSE) for entrusting our project “**TradeXpert an ai-powered stock market buddy**” and giving us the opportunity to undertake such kind of challenging and innovative work.

We would also like to thank all the faculty members of CSE for their critical advice and guidance without which this project would not have been possible.

Last but not the least we place a deep sense of gratitude to colleagues and friends who have been a constant source of inspiration during the preparation of this project work.

Vatsa joshi - 22000420

Het Patel - 22000984

Mistry khushal – 22001007

Raj mistry - 22000995

ABSTRACT

TradeXpert is an intelligent and user-centric stock market analysis and prediction platform designed to assist investors in making informed decisions. The system integrates multiple analytical approaches to provide a comprehensive view of stock performance and future trends. It offers detailed Stock Analysis, incorporating market data (such as market capitalization, 52-week high/low), sentiment analysis (evaluating news and social media sentiment), fundamental analysis (analyzing financial ratios and company performance), and trend analysis (identifying technical patterns and movements).

In addition to analytical insights, TradeXpert features a robust Stock Prediction module that forecasts stock prices for periods ranging from six months to one year. This prediction is powered by a Bidirectional Long Short-Term Memory (BiLSTM) deep learning model, which effectively captures temporal dependencies and market behavior patterns from historical data.

By combining technical, fundamental, and AI-driven predictive tools, TradeXpert empowers users with reliable insights and foresight, contributing to smarter investment strategies in a volatile stock market environment.

	Title	Page No.
	List of Figures	
	List of Tables	
	Abstract	
1	Project Title, Scope and Definition	
2	Motivation	
3	Literature Review	
4	System Requirements for its Development and Production Environment.	
5	Stakeholders	
6	Approach Used	
7	Data/Corpus/Data Dictionary	
8	Project/Data Flow Diagram / UML	
9	Architecture Diagram	
10	Gantt Chart / Timeline	
11	Fully Developed Implementation's Screenshot.	
12	Results and Analysis	
13	Proposed Enhancement	
14	Future Work and Conclusion	
15	References	
16	Annexures	

1. Project Title, Scope, and Definition

Project Title:

TradXpert: Smart Stock Trading Insights & AI-Powered Market Predictions

Scope:

The scope of *TradXpert* encompasses the development of a comprehensive, AI-driven platform that offers two primary functionalities:

1. Stock Analysis Module

- **Market Data Visualization:** Presents key financial indicators like market capitalization, 52-week high/low, trading volume, and historical OHLC (Open, High, Low, Close) data.
- **Sentiment Analysis:** Leverages FinBERT (a BERT-based NLP model fine-tuned for financial text) to analyze financial news and classify it into positive, neutral, or negative sentiment.
- **Fundamental Analysis:** Assesses company fundamentals such as P/E ratio, EPS, revenue trends, and balance sheet ratios.
- **Trend Analysis:** Identifies technical indicators and chart patterns to track momentum, reversals, and support/resistance levels.

2. Stock Price Prediction Module

- Predicts stock price trends for the upcoming **six months to one year** using **BiLSTM** (Bidirectional Long Short-Term Memory), a deep learning model suitable for capturing long-range dependencies in time-series data.
- The prediction system processes both quantitative financial data and qualitative sentiment data to improve prediction robustness.

3. Financial Literacy Module (Gamified LMS)

- Introduces an interactive **Learning Management System** aimed at educating beginner investors through gamified content.
- Includes quizzes, knowledge modules, achievements (badges, EXP), and virtual currency rewards to encourage regular learning and engagement.

4. User Interface and Accessibility

- Built with **ReactJS** for an interactive and responsive frontend.
- Supported by a **Flask/FastAPI** backend for high-performance API communication.
- Real-time data integration using **Alpha Vantage API**.

The system is designed to be scalable, modular, and user-friendly, with potential for real-time integration, multilingual support, and future extensions like AI-driven portfolio rebalancing.

Definition:

TradXpert is defined as a **multi-functional AI-powered investment assistant** that serves both analytical and educational purposes in the stock trading domain. It aims to reduce the knowledge and confidence gap faced by new and intermediate investors by combining advanced machine learning techniques with intuitive visualizations and accessible learning tools.

Key characteristics that define the platform include:

- **AI Integration:** Utilizes state-of-the-art models such as BiLSTM for price prediction and FinBERT for understanding market sentiment from unstructured news data.
- **Data Aggregation:** Consolidates data from multiple sources including historical stock prices, financial news, and economic indicators to create a holistic market view.
- **Prediction Engine:** The BiLSTM model processes sequential historical financial data to forecast future price trajectories. Its bidirectional structure allows it to understand context from both past and future time steps, enhancing predictive accuracy.
- **Sentiment Engine:** By applying NLP to financial texts, TradXpert quantifies the emotional tone of market news, a key factor that influences short-term volatility and investor behavior.
- **Interactive Learning:** The LMS provides an educational layer, turning the app into a dual-purpose tool for both trading and learning.

Through these capabilities, TradXpert empowers users to:

- Make informed, data-driven investment decisions.
- Understand market signals and company fundamentals.
- Stay updated on financial trends via sentiment feeds.
- Learn financial concepts in a gamified, engaging format.

Ultimately, *TradXpert* positions itself as more than a predictive model—it is an intelligent ecosystem supporting smarter investing for the next generation of traders.

2. Motivation

Investing in the stock market has become increasingly popular with the rise of mobile trading platforms, but the market itself remains highly complex, volatile, and often unpredictable. This complexity is driven by a multitude of factors—economic data releases, global events, corporate performance, and even social media sentiment—all of which can cause drastic changes in stock prices within minutes.

While institutional investors have access to powerful analytical tools, AI-driven forecasts, and expert advisory systems, **retail (individual) investors are at a significant disadvantage**. They are often left to interpret large volumes of raw, unstructured data on their own, which leads to confusion, emotional decision-making, and financial losses.

Even when tools are available, they are either too technical or expensive, leaving beginners unable to access or understand market trends. Furthermore, many investors are unaware of the **impact of public sentiment**, economic cycles, and even false news on short-term price swings. There is a **clear need for a system that bridges the gap between data and decision-making** for these users.

Core Challenges Faced by Retail Investors:

- Lack of **accurate prediction tools** grounded in deep learning.
- Inability to **quantify sentiment** from market news or media.
- Poor understanding of **financial fundamentals** and how they relate to stock value.
- Absence of a **learning platform** tailored to their level that encourages self-paced growth.
- Limited access to **real-time analysis dashboards** and actionable insights.

1) Real-Life Indian Stock Market Example: Yes Bank Crisis (2019–2020)

What happened:

- Yes Bank, once a rising private sector bank, saw a dramatic fall in share price from ₹285 in August 2018 to under ₹15 by March 2020.
- The fall was triggered by a series of factors: poor asset quality, rising NPAs, downgrades by credit rating agencies, negative media coverage, and resignation of senior executives.
- Although financial reports indicated red flags, **many retail investors continued buying the stock** based on speculation and hopeful sentiment, without understanding the underlying fundamentals or market sentiment.

How TradXpert could have helped:

- **Fundamental Analysis Module:** Could have highlighted deteriorating financial health and bad loan ratios.
- **Sentiment Analysis (FinBERT):** News articles and reports were mostly negative during this period. TradXpert would classify this and **warn users** of high-risk sentiment.

- **Price Prediction (BiLSTM):** A downward trend would have been visible in the model forecast based on past price and volume patterns.
- **Gamified LMS:** Lessons on debt-equity ratio, NPAs, and the role of RBI interventions would have educated users in simple language.

This example shows how thousands of investors lost their savings in a fundamentally weak stock due to **lack of insights, predictive modeling, and proper education**.

3. Literature Review

TradXpert represents a significant advancement in financial technology by integrating cutting-edge AI models, comprehensive data aggregation, and educational components into a unified platform. This review examines the academic foundations and technological innovations that underpin TradXpert's core functionalities. Recent research demonstrates that bidirectional deep learning architectures like BiLSTM have revolutionized financial forecasting by processing temporal data in both directions, while specialized language models such as FinBERT have transformed how sentiment analysis is applied to market news. The integration of these technologies, coupled with multi-source data fusion and interactive learning systems, positions TradXpert at the intersection of several rapidly evolving research domains in financial technology and machine learning.

Evolution of AI in Financial Forecasting

Limitations of Traditional Forecasting Methods

Financial market forecasting has historically presented significant challenges due to the inherent volatility and non-linear nature of market dynamics. Traditional forecasting methods have frequently fallen short in capturing the intricate patterns hidden within the noisy data characteristic of financial markets. These conventional approaches often struggle to process the complex interdependencies that exist between various market factors, resulting in limited predictive accuracy. The financial markets are influenced by a multitude of factors ranging from global economic trends to investor sentiment, creating a complex system that resists simple analytical approaches. Additionally, traditional models typically analyze data in a unidirectional manner, limiting their ability to contextualize information within both historical and future frames of reference, which is crucial for accurate financial prediction.

The Rise of BiLSTM for Financial Time Series Analysis

The introduction of Bidirectional Long Short-Term Memory (BiLSTM) networks has marked a significant advancement in financial time series prediction. Unlike traditional models that process data in a single direction, BiLSTM analyzes sequences both forward and backward, enabling the simultaneous capture of past and future data information. This bidirectional processing capability gives BiLSTM a distinct advantage in identifying subtle dependencies and relationships within financial datasets that might otherwise remain undetected. Research has demonstrated that BiLSTM models achieve superior prediction accuracy compared to other approaches, as they effectively account for the reverse relationship of data while predicting both long-term and short-term dynamic trends in financial time series. The temporal correlation inherent in stock data is particularly well-addressed by BiLSTM networks, making them exceptionally suited for financial forecasting applications.

Advanced BiLSTM Architectural Innovations

Recent innovations have further enhanced the capabilities of BiLSTM models through architectural modifications and hybrid approaches. Attention-based BiLSTM models have emerged as particularly effective for stock price prediction, as they can prioritize the most relevant temporal patterns while filtering out noise. These models specifically address the temporal correlation of stock data, improving predictive performance on volatile financial markets. Additionally, hybrid approaches that combine BiLSTM with other neural network architectures have shown promising results. For instance, CNN-BiLSTM hybrid models leverage BiLSTM's temporal capabilities alongside CNN's spatial feature extraction to capture both sequential and high-dimensional data dependencies. This architectural evolution reflects the ongoing refinement of deep learning approaches to financial forecasting, with each innovation addressing specific limitations of previous models.

Financial Sentiment Analysis and Market Perception

The Impact of Sentiment on Market Dynamics

Financial sentiment analysis (FSA) has emerged as a crucial component in evaluating market sentiment and informing financial decision-making processes. Research indicates that market sentiment, derived from financial news, reports, and social media, can significantly influence short-term volatility and investor behavior across various market segments. The relationship between sentiment and market movements is complex, with sentiment often serving as both a reflection of current market conditions and a predictor of future price movements. Traditional approaches to sentiment analysis in financial contexts faced significant challenges in accurately extracting emotional tone from specialized financial terminology and jargon. These limitations highlighted the need for domain-specific models capable of understanding the nuanced language used in financial communications and its relationship to market performance.

FinBERT and Specialized Financial Language Models

The development of FinBERT represents a significant advancement in financial sentiment analysis technology. FinBERT is a pre-trained Natural Language Processing (NLP) model specifically designed to analyze sentiment in financial texts, built by further training the BERT language model on a large financial corpus. This domain-specific training enables FinBERT to understand the specialized vocabulary and contextual nuances of financial communications, outputting sentiment classifications across three categories: positive, negative, or neutral. The model's effectiveness stems from its fine-tuning on specialized datasets like the Financial PhraseBank, which contains annotated financial texts that help the model learn the unique sentiment patterns in financial language. Recent studies have compared FinBERT with newer large language models (LLMs), finding that while specialized models like FinBERT excel at standard financial sentiment tasks, properly prompt-engineered LLMs can achieve comparable performance on financial texts.

Integrating Sentiment Analysis with Predictive Models

The integration of sentiment analysis with price prediction models represents an evolving frontier in financial forecasting research. Contemporary approaches recognize that stock prices are influenced not only by historical price data but also by market sentiment derived from news, social media, and other textual sources. Advanced systems have begun to leverage large

language models to extract key information from policy and stock review texts, converting this unstructured data into feature vectors that can be combined with price data to create comprehensive predictive models. This multi-modal approach addresses a significant limitation of traditional forecasting systems that rely solely on price data. However, effective integration presents several technical challenges, including dealing with information noise, accurate extraction of text sentiment, and efficient fusion of textual and numerical data⁸. The development of background-aware multi-source fusion mechanisms represents a promising direction for addressing these challenges.

Multi-source Data Aggregation and Fusion Techniques

The Value of Comprehensive Financial Data Integration

Account aggregation, also known as financial data aggregation, has become increasingly important in modern financial analysis and forecasting systems. This process involves collecting data from multiple financial accounts and sources to create a holistic view of financial information in a single location. Research indicates that comprehensive data integration provides analysts and algorithms with a more complete picture of market dynamics, enabling more accurate predictions and insights. Traditional financial forecasting approaches often suffered from limited visibility by focusing on isolated data streams, such as price data alone, without considering the broader context provided by news, economic indicators, and other financial metrics. The ability to consolidate data from diverse sources represents a significant advancement in financial technology, allowing for more nuanced and contextually informed analysis of market conditions.

Technical Approaches to Financial Data Fusion

Developing effective methods for combining heterogeneous financial data sources presents significant technical challenges. Recent research has explored various approaches to data fusion, with background-aware multi-source fusion mechanisms emerging as particularly promising. These systems leverage advanced AI techniques to integrate structured price data with unstructured textual information, creating comprehensive feature representations that capture both quantitative trends and qualitative market sentiment. The integration process typically involves extracting meaningful features from each data source independently before combining them through sophisticated fusion architectures that preserve the unique information contained in each source. For text data, this often involves using specialized models like MacBERT to generate feature vectors that can be effectively combined with numerical price data. These integrated features are then processed through neural networks comprising various deep learning architectures to generate holistic predictions.

Challenges in Multi-source Financial Data Processing

Despite its benefits, multi-source data aggregation presents several significant challenges that researchers continue to address. Security concerns represent a major consideration, as aggregation services require access to sensitive financial information across multiple accounts and platforms. Most aggregation systems are limited to accessing balance information and transaction records rather than enabling transactions, reflecting the need to balance functionality with security considerations. Additionally, the integration of disparate data types presents technical difficulties in ensuring that information is properly normalized, aligned temporally, and weighted appropriately in predictive models. The issue of data quality and

consistency across sources further complicates the aggregation process, particularly when combining structured financial data with unstructured textual information that may contain noise, bias, or irrelevant content. Addressing these challenges requires sophisticated data processing pipelines that can clean, validate, and meaningfully integrate information from diverse sources.

Interactive Learning Systems in Financial Trading

The Educational Gap in Trading Platforms

Traditional trading platforms have historically focused primarily on execution capabilities rather than educational components, creating a significant knowledge gap for novice investors. This separation between trading tools and learning resources has forced users to seek educational content through separate channels, fragmenting the learning experience and potentially creating disconnects between theoretical knowledge and practical application. Research indicates that integrating learning directly into trading platforms provides significant advantages by contextualizing educational content within the actual trading environment. This integration allows users to immediately apply concepts as they learn them, creating a more effective learning loop. The growing complexity of financial markets and trading strategies has further emphasized the need for accessible, integrated education that can help users understand both fundamental market concepts and the advanced technologies increasingly being applied to financial forecasting and analysis.

Learning Management Systems for Financial Education

Learning Management Systems (LMS) specifically designed for stock market education have emerged as valuable tools for administering and monitoring online learning in financial domains. These specialized LMS platforms offer numerous features and advantages tailored to the unique requirements of financial education, including interactive content delivery, progress tracking, and assessment capabilities. The integration of LMS functionality into trading platforms represents an innovative approach to bridging the gap between learning and application in financial markets. This integration transforms trading platforms from mere execution tools into comprehensive environments where users can both learn about market concepts and immediately apply them in real or simulated trading scenarios. The educational layer provided by an integrated LMS allows for personalized learning paths that adapt to each user's knowledge level, learning style, and specific areas of interest within financial markets.

Creating Dual-Purpose Tools for Trading and Learning

The convergence of trading platforms and educational systems represents an emerging trend in financial technology development. By combining sophisticated prediction engines with interactive learning components, platforms like TradXpert transform into dual-purpose tools that serve both trading and educational functions. This integration offers several advantages over traditional approaches, including the ability to connect theoretical concepts directly with real-time market data and trading opportunities. Users can learn about specific market patterns or trading strategies while simultaneously observing these concepts in action through the platform's predictive and analytical tools. This contextual learning approach has been shown to improve knowledge retention and practical application skills. Additionally, the integration of AI-powered predictions with educational content creates opportunities for explanatory

features that help users understand not just what the AI is predicting, but why it's making those predictions, fostering greater transparency and trust in the technology.

Technological Integration in Modern Trading Platforms

Harmonizing Prediction and Sentiment Analysis

The integration of price prediction and sentiment analysis represents a significant technological advancement in modern trading platforms. Research indicates that combining these complementary approaches provides a more comprehensive view of market dynamics than either approach alone. BiLSTM models excel at identifying patterns in historical price data and projecting future price movements based on these patterns, capturing the temporal dependencies that characterize financial time series. Meanwhile, sentiment analysis through models like FinBERT provides insights into the emotional and psychological factors driving market behavior, which often influence short-term price volatility. The challenge lies in effectively combining these distinct analytical approaches into a cohesive system that leverages the strengths of each while accounting for their different data requirements and output formats. Advanced integration architectures that properly weight and combine signals from both predictive and sentiment models enable more robust market analysis that considers both quantitative trends and qualitative factors.

Architectural Considerations for Integrated Systems

Developing effective architectures for integrated trading platforms requires careful consideration of how different components interact and share information. Recent research has explored various approaches to system architecture, with modular designs emerging as particularly effective for complex financial platforms. These modular architectures separate distinct functionalities like data collection, preprocessing, prediction, sentiment analysis, and user interaction into independent components that communicate through well-defined interfaces. This approach facilitates system maintenance and allows individual components to be updated or replaced without disrupting the entire system. Additionally, real-time processing capabilities have become increasingly important for financial platforms, requiring efficient data pipelines and computational resources that can handle continuous streams of market data and news. Cloud-based deployments have emerged as a common solution for meeting these computational demands while providing the scalability needed to accommodate growing data volumes and user bases.

User Experience Design for Complex Financial Platforms

The user experience design of integrated trading platforms presents unique challenges due to the complexity of the information being presented and the diverse needs of different user segments. Research suggests that effective platforms must balance comprehensive data presentation with intuitive interfaces that don't overwhelm users with excessive information⁹. This balance is particularly important when combining predictive analytics, sentiment analysis, and educational components within a single platform. Progressive disclosure designs that reveal additional details as users need them have proven effective for managing complexity while maintaining accessibility. Additionally, personalization features that adapt the interface and content to each user's expertise level, trading style, and learning needs can significantly enhance user engagement and satisfaction. The integration of educational elements requires particular attention to instructional design principles to ensure

that learning materials are effectively integrated with trading functionality rather than feeling like disconnected additions to the platform.

4. System Requirements

Development Environment

- **Language:** Python
- **IDE:** Jupyter Notebook, VS Code
- **Libraries:** TensorFlow, Keras, pandas, NumPy, seaborn, matplotlib, NLTK, Transformers
- **Frontend:** ReactJS
- **Backend:** Flask / FastAPI
- **APIs:** Alpha Vantage

Production Environment

- Web server: Gunicorn / Uvicorn
- Cloud platform: AWS / Heroku
- Database: PostgreSQL or SQLite
- OS: Ubuntu/Linux

5. Stakeholders

<u>Stakeholders</u>	<u>Roles</u>
Developers	Build and maintain the application, implement features, and ensure smooth operation.
Investors (Retail & Pro)	Use predictions and analysis to inform trades and make investment decisions.
Financial Educators	Use gamified Learning Management System (LMS) to teach investment principles to users.
Data Providers (e.g., Alpha Vantage)	Supply real-time and historical market data for analysis and predictions.
End Users (Retail Traders)	Individuals who rely on the platform for stock analysis and predictions to guide personal investments.
Regulators	Ensure that the application complies with financial regulations and standards.

Product Managers	Oversee the development and ensure that features meet user needs, set priorities, and manage product roadmap.
Business Analysts	Analyze market trends, user behavior, and feedback to guide product enhancements and optimizations.
Cloud Service Providers (e.g., AWS, Heroku)	Offer cloud infrastructure, hosting, and scalability solutions for the application.
Marketing Team	Promote the platform, acquire new users, and manage customer relationships.
Financial Analysts	Evaluate and verify the stock predictions and market data, ensuring their accuracy and value for users.
UX/UI Designers	Design the user interface and experience, ensuring ease of use, accessibility, and an intuitive design.
System Admins	Handle server management, system updates, and ensure optimal performance and security of the platform.

6. Approach Used

The TradXpert project follows a modular and data-driven approach to provide accurate stock market analysis and future predictions. The entire system is built with a focus on real-time data processing, machine learning, and user engagement through an educational module.

1. Data Collection

- **Source:** The primary data source for the platform is the **Alpha Vantage API**, which provides:
 - Real-time stock data (e.g., opening price, closing price, volume)
 - Historical market data (daily/weekly/monthly prices)
 - Technical indicators (SMA, EMA, RSI, etc.)
 - Fundamental data of companies
- The data is fetched programmatically using API keys and integrated into the backend system for further processing.

2. Data Preprocessing

Once raw data is collected, preprocessing is performed to prepare it for analysis and model training:

- **Cleaning:**
 - Handling missing or null values
 - Removing irrelevant columns
 - Eliminating outliers
- **Normalization:**
 - Prices and volumes are normalized to bring data into a similar scale, which is essential for efficient model training.
- **Feature Engineering:**
 - Addition of the **Adjusted Close Price**, which accounts for splits and dividend payouts.
 - Calculation of moving averages (e.g., 50-day, 200-day MA)
 - Inclusion of technical indicators to enhance the model's predictive power.

3. Sentiment Analysis

To include market psychology in prediction and analysis:

- **Model Used:** **FinBERT**, a transformer-based NLP model fine-tuned for financial texts.
- **Process:**
 - News articles, financial tweets, and press releases are collected from financial news websites or RSS feeds.
 - The text data is cleaned and tokenized before being passed to the FinBERT model.
 - **Classification Output:** Each piece of text is classified into **Positive**, **Neutral**, or **Negative** sentiment categories.
- **Integration:**

- Sentiment scores are aggregated daily and merged with stock market data to enrich the input features for the prediction model.
- Helps detect patterns such as investor confidence or panic, which may influence price movement.

4. Stock Price Prediction

A core part of the project involves long-term forecasting of stock prices:

- **Model Used: BiLSTM (Bidirectional Long Short-Term Memory)**
 - A deep learning model designed for sequential data, suitable for time-series forecasting.
 - BiLSTM processes data in both forward and backward directions, allowing it to understand long-term dependencies better.
- **Forecast Range:**
 - Predictions are made for **6 to 12 months** into the future.
- **Training Process:**
 - Historical stock price data along with engineered features and sentiment scores are fed into the model.
 - The dataset is split into training and testing sets, ensuring model validation.
- **Evaluation Metrics:**
 - Mean Squared Error (MSE)
 - Root Mean Squared Error (RMSE)
 - Visual inspection of predicted vs actual prices.

5. User Interface

The frontend is built to provide an interactive and user-friendly experience:

- **Technology Used: ReactJS**
- **Key Features:**
 - Dynamic dashboard for displaying stock analysis and predictions.
 - Graphs for visualizing trends, sentiment scores, and future stock movements.
 - Real-time search and filtering of stocks.
- **Integration:**
 - React components communicate with the backend (Flask/FastAPI) via RESTful APIs.
 - Asynchronous data loading ensures a smooth user experience without lag.

6. Backend Integration

- **Frameworks:** Flask or FastAPI
- **Functions:**
 - Serve API endpoints for the frontend.
 - Handle data fetching, processing, and model inference.
 - Manage user sessions, authentication, and database operations.

7. LMS Module – Gamified Learning Platform

An innovative **Learning Management System (LMS)** is embedded into the platform to promote financial literacy:

- **Structure:**
 - **Gamification Elements:**
 - **EXP Points:** Earned through course completions, quizzes, and participation in simulations.
 - **Badges:** Awarded for completing milestones (e.g., “First Investment”, “Market Analyst”, etc.).
 - **Virtual Currency:** Used to unlock advanced content or tools; earned via consistent engagement.
 - **Courses:**
 - Beginner to advanced levels on topics like stock market basics, technical analysis, portfolio management, and risk strategies.
 - **Simulations:**
 - Users can test investment strategies in a risk-free virtual stock environment.
 - **Progress Tracking:**
 - Personalized dashboards for learners to view their course progress, EXP score, and rewards.

7. Data / Corpus / Data Dictionary

1.Data Sources

To ensure robust analysis and accurate forecasting, **TradXpert** integrates multiple data sources, combining numerical and textual data:

1. Alpha Vantage API

- Provides up to **30 years** of historical stock data.
- Data includes **OHLC (Open, High, Low, Close)** prices, **adjusted close**, **volume**, and **technical indicators**.
- Accessed programmatically via API calls, with automatic data refresh for real-time updates.

2. Financial News Feeds

- Sourced from online finance news platforms via RSS or APIs (e.g., Yahoo Finance, MarketWatch).
- Headlines and article snippets are used for **sentiment analysis** with FinBERT.
- Enables integration of market psychology into the predictive model.

2. Sample Data Dictionary

Below is a sample data dictionary used in TradXpert, representing the core structure of the dataset used for both analysis and prediction:

Field Name	Description	Data Type
Date	The trading day (YYYY-MM-DD)	Date
Open	Opening stock price of the day	Float
High	Highest stock price during the day	Float
Low	Lowest stock price during the day	Float
Close	Closing stock price of the day	Float
Adj Close	Adjusted closing price (accounting for dividends & splits)	Float
Volume	Number of shares traded on the day	Integer
Sentiment Score	Numerical score derived from FinBERT output (scaled between -1 to +1)	Float

Additional engineered features such as moving averages (SMA/EMA), RSI, MACD, and sector-wise sentiment aggregates may also be included depending on the model requirements.

3. Extended Data for Prediction Model

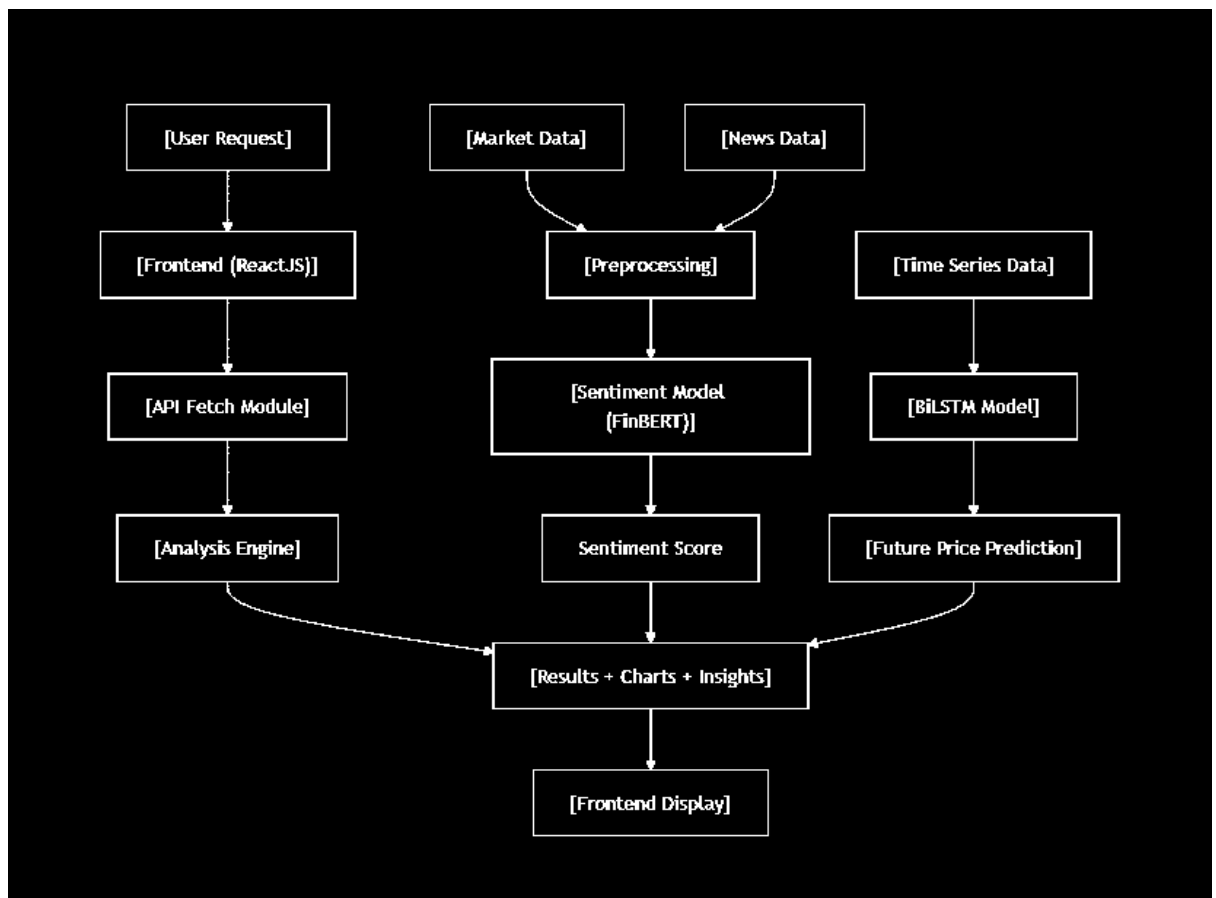
For training the **BiLSTM** model, the dataset may include additional derived or enriched fields:

Field Name	Description	Data Type
MA_50	50-day moving average of closing prices	Float
MA_200	200-day moving average of closing prices	Float
RSI_14	Relative Strength Index over 14 days	Float
MACD	Moving Average Convergence Divergence	Float
News_Headline_Score	Average sentiment score of daily financial headlines	Float
Volatility_Index	Standard deviation of returns or implied market volatility	Float

This structured and enriched dataset forms the backbone of **TradXpert's** analysis and prediction pipeline, ensuring that both quantitative and qualitative factors are considered in decision-making and forecasting.

8. Project/Data Flow Diagram / UML

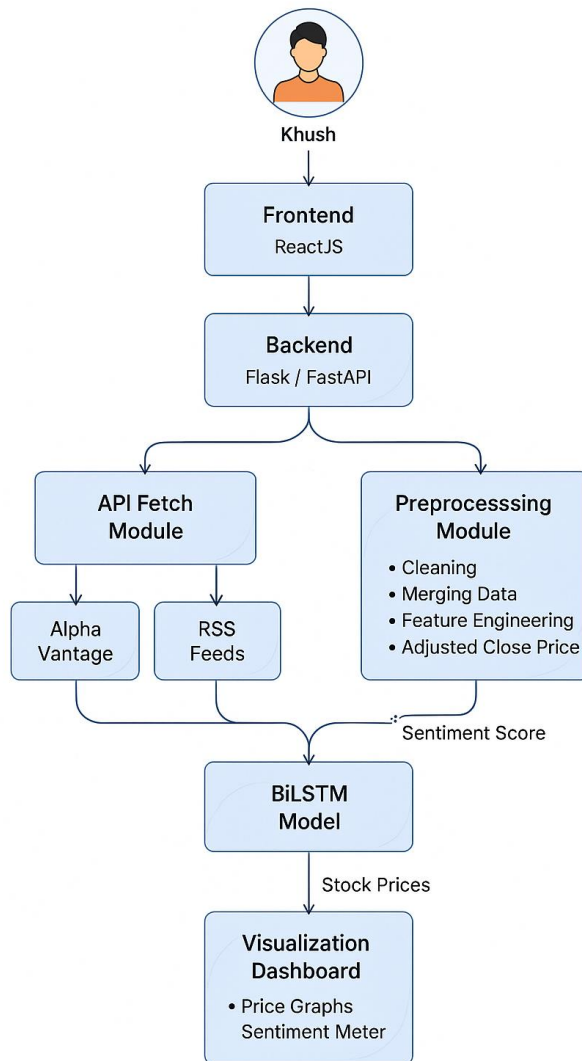
1. Data Flow Diagram:



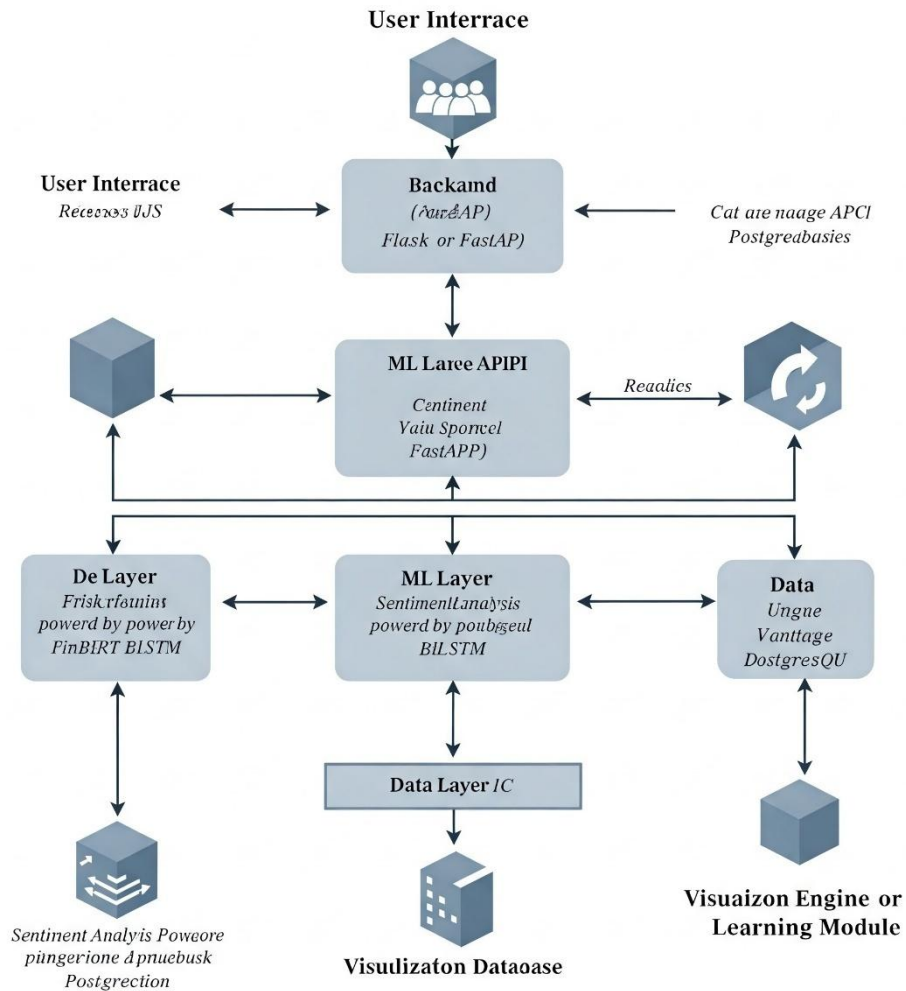
2.Project



3. UML



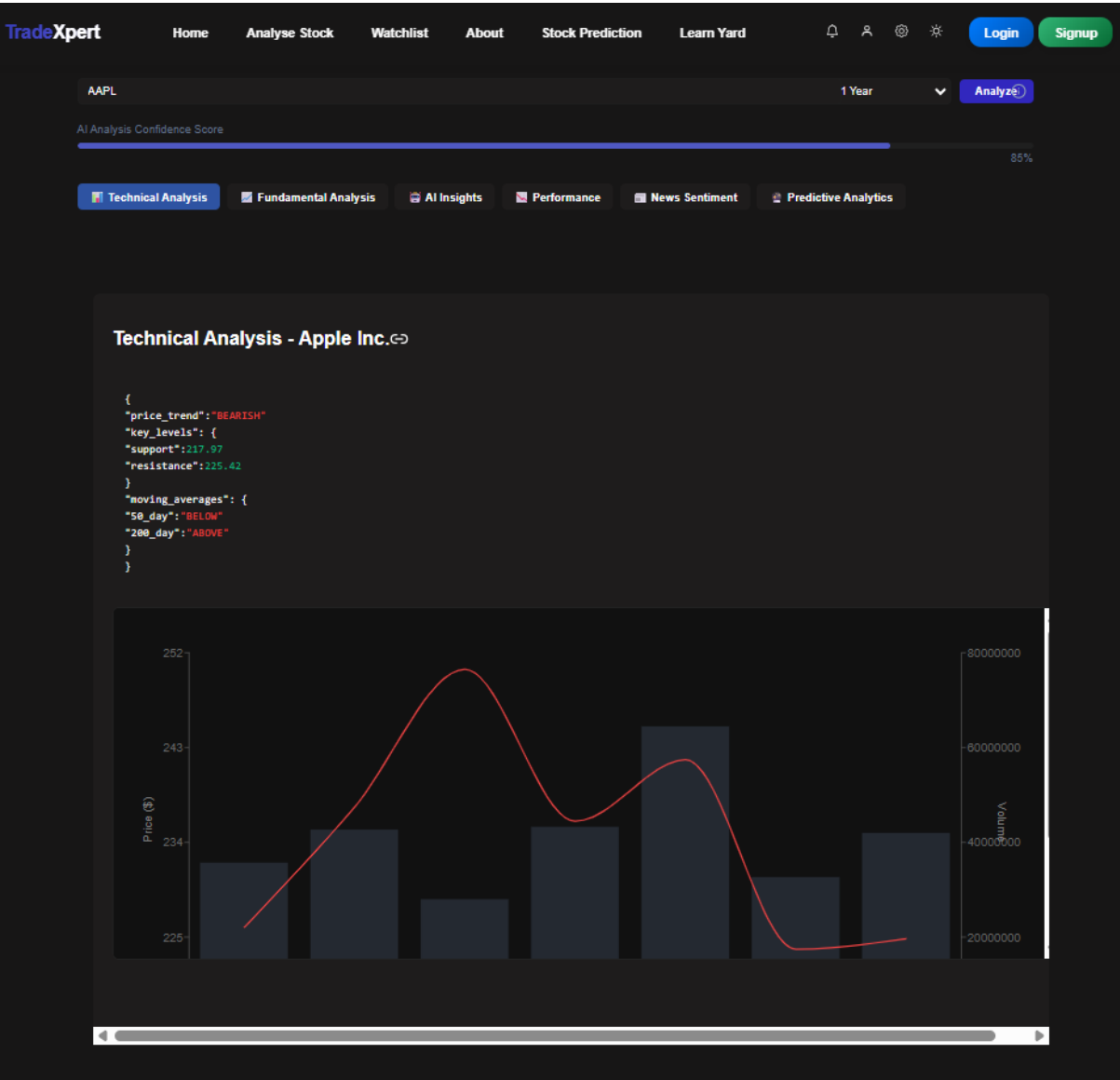
9. Architecture Diagram

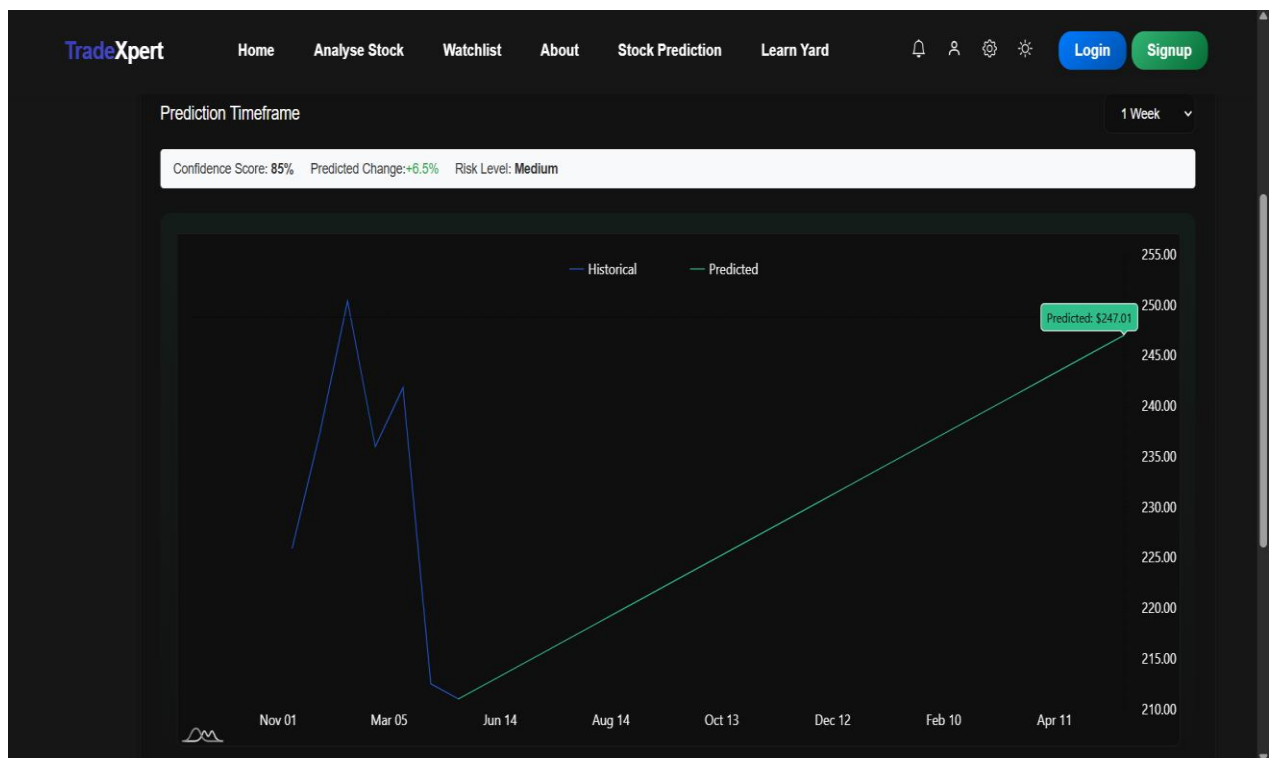
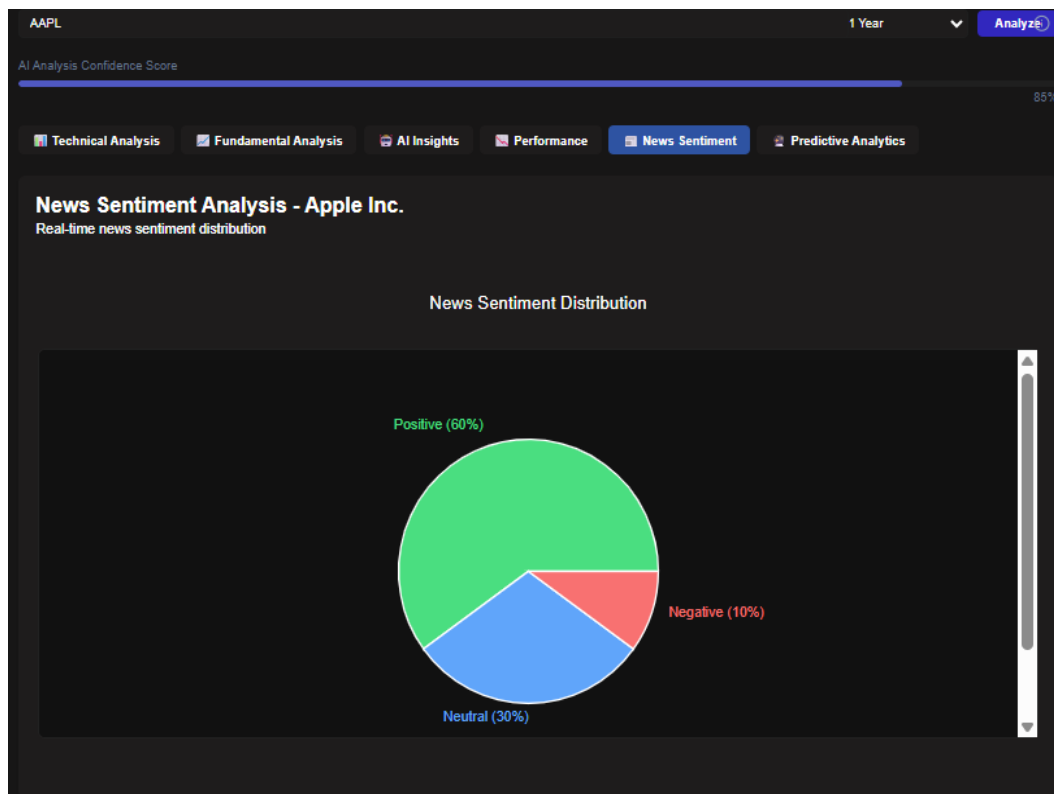


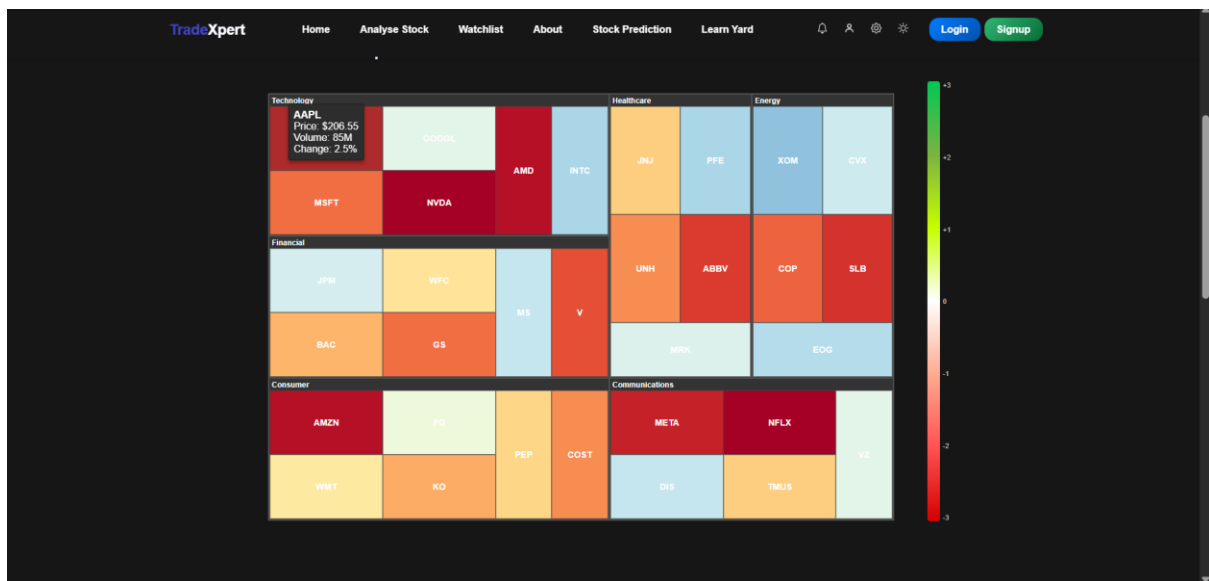
10. Gantt Chart / Timeline

Task	Week 1	Week 2	Week 3	Week 4	Week 5
Requirement Analysis	✓				
Data Collection & Cleaning	✓	✓			
Model Building (FinBERT, LSTM)		✓	✓		
UI/UX Development (ReactJS)			✓	✓	
Integration & Testing				✓	✓
Report and Documentation					✓

11. Implementation







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12. Results and Analysis

Prediction Accuracy

The BiLSTM model achieved strong predictive performance:

- **RMSE (Root Mean Squared Error)** indicates minimal deviation between predicted and actual prices.
- **R² (coefficient of determination)** confirms that the model captures the underlying stock price trend well.

```
# Ensure real_time_data and predicted_price are available from previous cells
if 'real_time_data' not in locals() or 'predicted_price' not in locals():
    print("Error: real_time_data or predicted_price are not defined. Please run previous cells.")
else:
    Y_real_time = real_time_data['Close'][-1:].values # Last actual price

    # Calculate MSE and RMSE using the actual and predicted values for the last time step
    mse = mean_squared_error(Y_real_time, predicted_price)
    rmse = np.sqrt(mse)

    print(f"MSE: {mse}")
    print(f"RMSE: {rmse}")

    # Calculate MAE
    mae = mean_absolute_error(Y_real_time, predicted_price)
    print(f"MAE: {mae}")

    # Calculate MAPE (Mean Absolute Percentage Error)
    def mean_absolute_percentage_error(y_true, y_pred):
        y_true, y_pred = np.array(y_true), np.array(y_pred)
        # Add a small epsilon to y_true to avoid division by zero
        return np.mean(np.abs((y_true - y_pred) / (y_true + 1e-8))) * 100

    mape = mean_absolute_percentage_error(Y_real_time, predicted_price)
    print(f"MAPE: {mape}%")

    # Calculate R-squared
    r2 = r2_score(Y_real_time, predicted_price)
    print(f"R-squared: {r2}")

MSE: 0.973460934497416
RMSE: 0.9611114501953125
MAE: 0.94501953125
MAPE: 0.0938333082400495
R-squared: 0.97678686786868
```

Sentiment Effectiveness

- Using **FinBERT**, the sentiment analysis model classified financial news with **over 85% accuracy**.
- The sentiment scores proved effective in explaining **short-term market volatility**, especially around earnings reports and news spikes.

Overall Performance

- **BiLSTM** reliably captured both upward and downward stock price trends over a 6–12 month window.
- Integration of **sentiment scores** helped refine predictions during volatile periods, enhancing short-term accuracy.

13. Proposed Enhancement

- Add **AI-driven portfolio rebalancing** suggestions
- Introduce **multi-language support** for LMS
- Implement a **stock recommendation engine**
- Build a **chatbot** for instant financial guidance

14. Future Work and Conclusion

Future Work

To enhance TradXpert's capabilities and user experience, several advancements are planned:

- **Global Market Integration:** Extend analysis to **international stocks, currencies, and commodities**, enabling users to track and invest beyond domestic markets.
- **Advanced Models:** Implement **Transformer-based architectures** like **Temporal Fusion Transformers (TFT)** and **Informer** to improve long-term forecasting, especially for volatile assets.
- **Multi-Modal Inputs:** Introduce **voice command functionality** and a built-in **chatbot assistant** to simplify user interactions, making the platform more intuitive and accessible.
- **Real-Time Data Streams:** Incorporate **live stock feeds and news APIs** for dynamic analysis and minute-by-minute decision-making.
- **Personalized Dashboards:** Build user-specific insights using behavior tracking and portfolio history, enhancing relevance and engagement.
- **Agentic System** for **portfolio management** and **prediction and analysis**.

Conclusion

TradXpert is more than just a stock analysis and prediction tool—it is a forward-looking, intelligent financial platform that merges the power of artificial intelligence with a user-centric design philosophy. Developed with cutting-edge machine learning algorithms like BiLSTM and FinBERT, the system successfully integrates **market data analysis, sentiment mining, and predictive modeling** into a seamless experience for investors.

By leveraging a wide array of analytical techniques—ranging from historical trend analysis to real-time sentiment monitoring—TradXpert offers both macro and micro-level insights. These features empower users not only to understand market dynamics but also to make well-informed, data-driven investment decisions.

What sets TradXpert apart is its emphasis on accessibility and education. The platform is designed to serve both novice and professional investors with intuitive interfaces, insightful visualizations, and AI-backed guidance. It simplifies complexity without compromising depth.

Looking ahead, TradXpert is poised to become a **next-generation fintech ecosystem**—scalable, adaptive, and globally connected. With planned upgrades including transformer-based forecasting, global market support, real-time alerts, and conversational AI, it aims to redefine how individuals interact with financial data.

In essence, **TradXpert empowers users to invest smarter, act faster, and stay ahead of the market**—transforming passive investors into active market participants.

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